

Amplification Must Be Paid: A Conditional Source-Ledger Framework for Navier–Stokes

Joseph Roy

June 2026

Abstract

I propose a scale-critical accounting principle for three-dimensional Navier–Stokes dynamics: persistent amplification must be paid by an identifiable channel. The guiding intuition is that constraints are not created or destroyed by a blowup scenario; they are redistributed among source, flux, strain, renewal, quotient, separation, and residual channels. The proposed ledger tries to make that redistribution visible.

The framework is built around the positive vortex-stretching density $\sigma_+ = (\omega \cdot S\omega)_+$ and a scale-critical denominator ledger $D(r) = \mathbf{1}^T X(r)$. The source ledger is formulated under a local source-finiteness or truncation hypothesis,

$$P(r_0) = r_0 \iint_{Q_{r_0}(z_0)} \sigma_+ \, dx \, dt < \infty,$$

which is automatic in smooth pre-singular regions but is not automatic in the bare Leray–Hopf energy class.

This manuscript does not claim a complete proof of Navier–Stokes regularity. It gives a stand-alone statement of the amplification-payment program, defines the live ledger rows, records partial closure mechanisms, states the conditional regularity theorem with its constant gap, reports adversarial numerical tests only as mechanism tests, and lists the remaining theorem gates.

1 Introduction

The Navier–Stokes equations have conservation and dissipation structures, but possible singularity formation is a problem of amplification. The guiding idea of this manuscript is that amplification should not be free. If vorticity, strain, boundary flux, annular leakage, or coherent residuals persist at a scale-critical level, then the dynamics should generate a visible payment in a finite ledger.

The proposed law is not a new PDE. It is a derived accounting principle:

amplification must be paid.

This paper is designed to stand alone as the mechanism paper: it defines the ledger, states what is proved conditionally, explains what has been tested, and makes the remaining proof gates explicit. The downstream technical gates K3 through K7 are isolated in the companion note `technical_gates_k3_k7.pdf`.

2 The Proposed Law

The intuition behind the proposed law is simple: constraints cannot be created or destroyed for free; they can only be redistributed. If this intuition is right, then a Navier–Stokes growth scenario cannot create persistent scale-critical amplification without sending the cost somewhere visible. The ledger below is one way to make that intuition falsifiable: amplification must route to source, boundary, tail, renewal, quotient redundancy, separation, coherent residual, or an explicitly unresolved row.

Principle 1 (Amplification payment principle). *Persistent scale-critical amplification in three-dimensional incompressible Navier–Stokes should generate an identifiable payment channel. In the source-ledger formulation, this takes the schematic form*

$$\text{Amplification}(r) \leq \text{Payment}(r) + o_r(1), \quad (1)$$

where $\text{Payment}(r)$ is a finite ledger of source, boundary, tail, strain, renewal, quotient, separation, and residual charges. In the source-ledger formulation below, the amplification and payment terms are represented through $P(r)$, $D(r)$, and the final physical ledger.

Remark 1 (Conjectural status). *Principle 1 is a conjectural organizing law. The paper does not assert (1) as an unconditional theorem. The conditional theorem in Section 6 states exactly which remaining gates would turn the law into a regularity contradiction.*

Throughout, $o_r(1)$ denotes a quantity tending to zero as $r \downarrow 0$ along the stated sequence of scales, with all fixed ledger parameters held constant. When the proof uses selected scales $r_n \downarrow 0$, I also write $o_n(1)$.

Standalone versus technical-ledger dependencies

This manuscript is designed to be readable without the full technical ledger, but it does not make every ledger estimate self-contained. It proves or records the following ingredients:

1. the conditional amplification-payment closure theorem;
2. the source-doubling scale-selection lemma;
3. fixed-dilation and bounded-macroscopic-parent tail payment;
4. the empirical attack summaries as mechanism tests.

It imports the following technical-ledger inputs:

1. denominator-to-CKN compatibility;
2. K3 quotient-or-charge for the final physical ledger;
3. K4 forward absorption of the final physical ledger;
4. K5 reverse matrix constants, including packed boundary-row constants;
5. K5B high-strain transport rigidity;
6. K6 spectral-radius and loop-gap certification.

Thus the paper should be read as a conditional program note: it isolates the finite list of theorem gates rather than claiming that all gates are already proved here.

3 Source Ledger and Denominator

Fix a candidate point $z_0 = (x_0, T)$ and set

$$Q_r(z_0) := B_r(x_0) \times (T - r^2, T).$$

Let $S = (\nabla u + \nabla u^T)/2$, $\omega = \nabla \times u$, and

$$\sigma_+ := (\omega \cdot S\omega)_+.$$

The source ledger is

$$P(r) := r \iint_{Q_r(z_0)} \sigma_+ dx dt. \quad (2)$$

Remark 2 (Source-finiteness input). *For fixed parent cylinders used in the ledger, I assume or import*

$$P(r_0) = r_0 \iint_{Q_{r_0}(z_0)} \sigma_+ dx dt < \infty.$$

In smooth pre-singular regions this follows from local smoothness. For suitable weak solutions it is a serious additional source-ledger hypothesis, or else it must be replaced by a truncation/renormalization version of the framework.

Let ϕ_r be a standard nonnegative cutoff adapted to $Q_{2r}(z_0)$, equal to one on $Q_r(z_0)$, with

$$|\nabla \phi_r| \lesssim r^{-1}, \quad |\partial_t \phi_r| + \nu |\Delta \phi_r| \lesssim r^{-2}.$$

The live denominator is

$$D(r) := \mathbf{1}^T X(r), \quad X(r) := \begin{pmatrix} \mathcal{E}_\omega(r) \\ \mathcal{B}_\omega(r) \\ \mathcal{D}_{\text{ARR}}(r) \\ \mathfrak{B}_{\text{strain}}(r) \\ M_{\text{deact}}(r) \\ \mathcal{E}_\nu^{\text{coh},+}(r) \end{pmatrix}. \quad (3)$$

All rows of $D(r)$ are required to be nonnegative, scale-critical, and stable under the scale subsequences used in Theorem 1. The normalizations used here are the following.

1. Vorticity energy/dissipation.

$$\begin{aligned} \mathcal{E}_\omega(r) &:= r \operatorname{ess\,sup}_{t \in (T-4r^2, T)} \int_{B_{2r}} |\omega|^2 \phi_r^2 dx \\ &\quad + \nu r \iint_{Q_{2r}} |\nabla \omega|^2 \phi_r^2 dx dt. \end{aligned}$$

2. Boundary/collar flux.

$$\mathcal{B}_\omega(r) := \mathcal{B}_\omega^{\text{sh}}(r) + \mathcal{B}_{\omega, \lambda_0}^{\text{pack}}(2r),$$

where

$$\mathcal{B}_\omega^{\text{sh}}(r) := r \iint_{Q_{2r} \setminus Q_r} |\omega|^2 \Theta_{\phi_r} dx dt$$

and

$$\Theta_{\phi_r} := |\partial_t \phi_r| + \nu |\Delta \phi_r| + |u - \bar{u}_{B_{2r}}(t)| |\nabla \phi_r|.$$

3. Annular renewal/tail reservoir.

$$\mathcal{D}_{\text{ARR}}(r) := S_{\text{ARR}}^{\text{tail}}(r) + o_r(1),$$

with

$$S_{\text{ARR}}^{\text{tail}}(r) := \sum_{k \geq 0} 2^{-k} 2^{k+1} r \iint_{\mathcal{A}_k(r)} \sigma_+ \chi_k^{\text{tail}} dx dt.$$

Here

$$\mathcal{A}_k(r) := Q_{2^{k+1}r}(z_0) \setminus Q_{2^k r}(z_0)$$

inside the chosen parent cylinder.

4. Source-weighted strain demand.

$$\mathfrak{B}_{\text{strain}}(r) := r \iint_{\mathcal{U}_r} \sigma_+ S_r \phi_r^2 dx dt, \quad S_r := r^2 |S_{\text{loc}}|.$$

The raw term $r \iint \sigma_+ |S_{\text{loc}}| \phi_r^2$ is excluded as supercritical.

5. Source deactivation.

$$M_{\text{deact}}(r) := r \iint_{\cup_j \mathcal{T}_{r,j} \cap Q_r} \sigma_+ \mathbf{1}_{\text{deact}} \phi_r^2 dx dt + o_r(1),$$

where $\mathbf{1}_{\text{deact}}$ marks promoted source that fails the persistence/deactivation test.

6. Coherent viscous residual.

$$\mathcal{E}_{\nu}^{\text{coh},+}(r) := r \iint_{Q_{2r}} \mathcal{R}_{\nu, \text{coh}}^+ dx dt.$$

This is the positive coherent part of the viscous/projection residual left by the localized vorticity ledger after source, boundary, and tail rows are separated.

Technical-ledger symbols

The following symbols are not new proof inputs in this paper; they are the named placeholders used by the technical ledger.

- S_{loc} is the locally normalized strain tensor used in the source-weighted strain row, after subtracting the harmless large-scale or harmonic component fixed by the cutoff convention.
- $\mathcal{T}_{r,j}$ are promoted finite-thickness source tubes generated by the primary-source promotion theorem.
- χ_k^{tail} is the tail-membership cutoff assigning promoted source mass in the k -th annular parent shell to the ARR row.
- $\mathbf{1}_{\text{deact}}$ marks promoted source packets that fail the persistence/renewal test and are therefore routed to M_{deact} .
- $\mathcal{R}_{\nu, \text{coh}}^+$ is the positive coherent part of the viscous/projection residual after source, boundary, tail, and strain terms have been separated in the localized vorticity ledger.

- Quotient redundancy means that overlapping promoted packets are identified by chains of significant-overlap pairs whose final physical distance is at most the quotient threshold δ .

The companion technical ledger must provide exact constructions and constants for these objects. This paper records where each object enters the conditional closure.

The packed boundary contribution is

$$\mathcal{B}_{\omega, \lambda_0}^{\text{pack}}(R) := \sup_{\substack{0 < a < b \\ \lambda_0 b \leq R}} \int_a^b \mathcal{B}_{\omega}^{\text{adv}}(s, \lambda_0) \frac{ds}{s}, \quad (4)$$

where

$$\mathcal{B}_{\omega}^{\text{adv}}(s, \lambda_0) := s \iint_{Q_{\lambda_0 s} \setminus Q_s} |u - \bar{u}_{B_{\lambda_0 s}}(t)| |\omega|^2 |\nabla \phi_{s, \lambda_0}| \phi_{s, \lambda_0} \, dx \, dt.$$

This packed row is essential for the good-scale null-source result; replacing it by a single shell reopens the boundary contraction problem.

4 Physical Charges and Quotient Ledger

The final physical ledger is schematically

$$\mathcal{F}_{\text{phys}}^{\text{final}} := \mathcal{F}_{\text{phys}} + R_{\text{osc}}^{\text{fam}}, \quad (5)$$

where $\mathcal{F}_{\text{phys}}$ includes source-weighted shape, renewal, spread/separation, activity, boundary, tail, and coherent-residual coordinates. The oscillation/twist coordinate is explicit because large accumulated strain need not imply large net shape growth.

For promoted branches i, j , the final physical distance has the form

$$d_{\text{phys}}^{\text{final}}(i, j) := \max\{d_{\text{phys}}(i, j), w_{\text{osc}} \Phi_{\text{osc}}(i, j)\}. \quad (6)$$

The quotient relation $i \sim_{\delta} j$ identifies branches connected by significant-overlap chains with

$$d_{\text{phys}}^{\text{final}} \leq \delta.$$

The nonredundant edge set is

$$E_{\text{nonred}}^{\text{final}} := \{(i, j) : i \not\sim_{\delta} j, m_{ij} > 0\},$$

where m_{ij} is the source-overlap edge mass. The K3 target is

$$\sum_{(i,j) \in E_{\text{nonred}}^{\text{final}}} m_{ij} \geq c_m^{\text{final}} \eta P(r), \quad (7)$$

$$\sum_{(i,j) \in E_{\text{nonred}}^{\text{final}}} m_{ij} d_{\text{phys}}^{\text{final}}(i, j) \leq C_*^{\text{final}} P(r) \mathcal{F}_{\text{phys}}^{\text{final}}(r). \quad (8)$$

Together these imply

$$\mathcal{F}_{\text{phys}}^{\text{final}}(r) \geq c_{\text{final}}(\eta), \quad c_{\text{final}}(\eta) := \frac{\delta c_m^{\text{final}}}{C_*^{\text{final}}} \eta. \quad (9)$$

5 Regularity Bridge

The final regularity step is imported from standard local regularity theory, but the normalization must be explicit. Define the mean-subtracted CKN quantity

$$\mathcal{K}_{\text{CKN}}^{\text{ms}}(r; z_0) := r^{-2} \iint_{Q_r(z_0)} \left(|u - \bar{u}_{B_r}(t)|^3 + |p - \bar{p}_{B_r}(t)|^{3/2} \right) dx dt. \quad (10)$$

Proposition 1 (Imported denominator-to-CKN bridge). *Assume the denominator compatibility used in the technical ledger:*

$$D(r_n) \rightarrow 0 \implies r_n \iint_{Q_{2r_n}} |\omega|^3 dx dt \rightarrow 0$$

and the harmless cutoff, harmonic, and pressure-projection remainders vanish. Then

$$D(r_n) \rightarrow 0 \implies \mathcal{K}_{\text{CKN}}^{\text{ms}}(r_n; z_0) \rightarrow 0.$$

Consequently, for sufficiently large n , the CKN epsilon-regularity criterion applies at scale r_n , so z_0 is regular.

Imported mechanism. The bridge uses local Gagliardo–Nirenberg interpolation for vorticity, local div–curl/Biot–Savart control of the mean-subtracted velocity, local pressure decomposition with Calderon–Zygmund estimates, and the mean-subtracted Caffarelli–Kohn–Nirenberg epsilon criterion. No source-packing argument is used in this final imported step. $\square$

6 Conditional Closure Algebra

The theorem below is an algebraic closure theorem. Its role is to show that the listed analytic inputs are sufficient for regularity. It does not prove the imported K3–K7 inputs.

Theorem 1 (Algebraic conditional amplification-payment closure). *Let z_0 be a candidate singular point. Suppose the following inputs hold along a scale sequence $r_n \downarrow 0$:*

1. **Persistent source and promotion:**

$$P(r_n) \geq p_* > 0$$

and a finite-thickness promoted source family captures a fixed fraction $\eta > 0$.

2. **Quotient-or-charge:** *either the promoted family is in the nonredundant charge branch and*

$$\mathcal{F}_{\text{phys}}^{\text{final}}(r_n) \geq c_{\text{final}}(\eta)$$

holds, or the quotient-redundant branch has already been routed to a null-source/good-scale alternative that gives regularity.

3. **Forward absorption:**

$$P(r_n)\mathcal{F}_{\text{phys}}^{\text{final}}(r_n) \leq C_{\text{forw}}^{\text{final}}D(r_n) + o_n(1).$$

4. **Reverse matrix absorption:**

$$X(r_n) \leq AX(r_n) + bP(r_n) + o_n(1),$$

where $A \geq 0$, $b \geq 0$, and $\rho(A) < 1$. Here $o_n(1)$ in the vector inequality denotes a vector whose components tend to zero; inequalities are interpreted componentwise. Set

$$C_0 := \mathbf{1}^T(I - A)^{-1}b.$$

5. **Strict loop gap:**

$$C_{\text{forw}}^{\text{final}}C_0 < c_{\text{final}}(\eta).$$

6. **CKN bridge:** *Proposition 1 applies to the chosen denominator D .*

Then z_0 is regular.

Proof. If the quotient-redundant alternative has already been routed to a null-source/good-scale regularity conclusion, there is nothing further to prove. Otherwise the contradiction sequence is in the nonredundant charge branch. By quotient-or-charge and forward absorption,

$$c_{\text{final}}(\eta)P(r_n) \leq P(r_n)\mathcal{F}_{\text{phys}}^{\text{final}}(r_n) \leq C_{\text{forw}}^{\text{final}}D(r_n) + o_n(1).$$

Since $\rho(A) < 1$, the reverse matrix inequality gives

$$X(r_n) \leq (I - A)^{-1}bP(r_n) + o_n(1),$$

and therefore

$$D(r_n) = \mathbf{1}^T X(r_n) \leq C_0 P(r_n) + o_n(1).$$

Substitution yields

$$c_{\text{final}}(\eta)P(r_n) \leq C_{\text{forw}}^{\text{final}}C_0P(r_n) + o_n(1).$$

Because $C_{\text{forw}}^{\text{final}}C_0 < c_{\text{final}}(\eta)$ and $P(r_n) \geq p_* > 0$, this is impossible for large n . Thus the persistent-source nonredundant charge branch is ruled out, while the quotient branch is handled by the routed quotient/null-source alternative. Equivalently, the loop forces $P(r_n) \rightarrow 0$ and then $D(r_n) \rightarrow 0$; Proposition 1 gives regularity. $\square$

Remark 3 (Weighted spectral certificate). *The condition $\rho(A) < 1$ may be certified without eigenvalue calculation. If $w_i > 0$ and*

$$q_A(w) := \max_i \frac{1}{w_i} \sum_j A_{ij}w_j < 1,$$

then $\rho(A) \leq q_A(w) < 1$. A usable reverse constant is

$$C_0^{\text{cert}}(w) := \frac{(\sum_i w_i) \max_i (b_i/w_i)}{1 - q_A(w)}.$$

The certified loop gap is

$$C_{\text{forw}}^{\text{final}}C_0^{\text{cert}}(w) < c_{\text{final}}(\eta).$$

7 Partial Closure Results

The following mechanisms are currently the most stable pieces of the program. They are stated here in simplified form; their detailed proofs and proof obligations are reflected in the technical ledger.

Proposition 2 (Good-scale null-source regularity). *If the packed boundary row (4) is part of $D(r)$ and*

$$\limsup_{r \downarrow 0} P(r) = 0,$$

then there exists $r_j \downarrow 0$ such that $D(r_j) \rightarrow 0$, hence z_0 is regular by Proposition 1.

Mechanism. The localized vorticity identity separates the collar flux into a source-active part and a no-source advective part. The source-active part is paid by P . The no-source part obeys a logarithmic packing estimate:

$$\int_r^{\Lambda r} \mathcal{B}_\omega^{\text{adv, nosrc}}(s, \lambda_0) \frac{ds}{s} \leq C_{\text{avg}} D(\Lambda \lambda_0 r) + o_r(1).$$

Pigeonholing selects $s \in [r, \Lambda r]$ with contraction coefficient

$$\theta_* = \theta_{\text{cut}}(\Lambda, \lambda_0) + \frac{C_{\text{avg}}}{\log \Lambda} < 1$$

for large enough Λ . Iteration gives a good scale with $D \rightarrow 0$. $\square$

Proposition 3 (Source-doubling scale selection). *If $\limsup_{r \downarrow 0} P(r) > 0$, then for every fixed $\Lambda_{\text{par}} > 1$ and $C_{\text{dbl}} > 1$ there exist $r_n \downarrow 0$ and $p_* > 0$ such that*

$$P(r_n) \geq p_*, \quad P(\Lambda_{\text{par}} r_n) \leq C_{\text{dbl}} P(r_n).$$

Proof. Assume otherwise. Choose $p_* < \limsup P$. If every r with $P(r) \geq p_*$ satisfies

$$P(\Lambda_{\text{par}} r) > C_{\text{dbl}} P(r),$$

then iteration gives

$$P(\Lambda_{\text{par}}^m r) > C_{\text{dbl}}^m p_*$$

while $\Lambda_{\text{par}}^m r$ remains in a fixed finite parent cylinder. This contradicts local source-ledger finiteness on that parent cylinder as $m \rightarrow \infty$. $\square$

Proposition 4 (Fixed-dilation promoted-tail payment). *Assume promoted-tail slices obey bounded overlap in a parent cylinder:*

$$\sum_{j,k} \mathbf{1}_{\mathcal{T}_{j,k}^{\text{tail}}(r)} \leq N_{\text{tail}} \mathbf{1}_{Q_{\Lambda_{\text{par}} r}(z_0)} + o_r(1).$$

Along source-doubling radii,

$$r_n \sum_{j,k} \iint_{\mathcal{T}_{j,k}^{\text{tail}}(r_n)} \sigma_+ dx dt \leq C_{\text{tail}}^{\text{fp}} P(r_n) + o_n(1).$$

Proposition 5 (Bounded macroscopic-parent tail smallness). *Assume promoted-tail slices obey bounded overlap in a fixed parent cylinder:*

$$\sum_{j,k} \mathbf{1}_{\mathcal{T}_{j,k}^{\text{tail}}(r_n)} \leq N_{\text{tail}} \mathbf{1}_{Q_{r_0}(z_0)} + o_n(1),$$

with $P(r_0) < \infty$. Then

$$r_n \sum_{j,k} \iint_{\mathcal{T}_{j,k}^{\text{tail}}(r_n)} \sigma_+ dx dt = o_n(1).$$

Proof. Bounded overlap gives a parent source bound. Since

$$r_n \iint_{Q_{r_0}(z_0)} \sigma_+ dx dt = \frac{r_n}{r_0} P(r_0) \rightarrow 0,$$

the fixed macroscopic parent contribution is negligible. □

8 Adversarial Mechanism Tests

The empirical layer is an attack layer. It checks whether the proposed payment channels appear in cached turbulent data and adversarial finite-radius audits. It does not prove any continuum inequality.

Glossary of computational labels

The labels $c413$, $c309$, and $c185$ denote cached promoted-source candidate families extracted from Johns Hopkins Turbulence Database diagnostics [10, 11]. The labels $D3$, $D5$, $D6$ denote denominator variants used in the tail-denominator ladder audit: $D5$ is the local-plus-observed-annular capacity used by that audit, while $D3$ and $D6$ require same-row moving-tube or velocity-pressure CKN data that were unavailable for those rows. The tag “ARR $c185$ final81” denotes the annular-renewal reservoir audit for the $c185$ candidate at support size 81.

Attack or audit		Observed result	Interpretation
Near-degenerate split	same-parent	For $M = 1, 4, 8, 16, 32, 64$, same-parent sub-tubes in cached $c413/c309$ families collapse under the physical quotient; captured source remains $\eta = 1$.	Artificial label multiplicity is quotient-redundant.
Renewal-cascade jitter	cross-parent	High capture persists, but post-quotient tube growth remains large; available physical cost stays positive in completed $c413/c309$ rows.	The attack stresses physical charge and renewal rather than producing a low-cost hidden packet.
Tail denominator ladder		In 15 rows the best available denominator was $D5$, while $D3, D6$ were unresolved. ¹	Arbitrary tail payment is not automatic; missing denominator families must be named, not filled by proxy.
ARR $c185$ final81 attribution		Renewal-exposure correction moves the ARR pressure ratio from about 1.022 to 0.967.	Renewal exposure is a meaningful accounting channel in this audit.
Coherent residual attribution		Finite-radius $c413$ rows route to renewal/deactivation or signed damping, not to a dangerous coherent positive source.	Supports the residual-routing taxonomy, but does not prove the coherent row.

Remark 4 (Evidence fence). *The tests are falsification and mechanism checks. They do not prove source-ledger finiteness, $K3$ quotient-or-charge, $K4$ forward absorption, $K5$ reverse matrix closure, $K5B$ high-strain rigidity, or the $K6$ loop gap. Their value is that they refuse to hide missing continuum quantities: unavailable rows remain marked as unresolved.*

¹The public package includes derived cached rows and notebook checks, not raw JHTDB dumps or the full extraction pipeline. The underlying audit recorded best ratios ranging from about 1.27 to 22.56.

9 Remaining Theorem Gates

The program is useful only if the remaining gates are explicit. The downstream technical gates K3 through K7 are isolated in `technical_gates_k3_k7.pdf`; the list below is the short dependency ledger used by this paper.

1. **Source-ledger finiteness or truncation.** Formulate the source ledger in the intended solution class, or replace P by a truncation with a stable limiting argument.
2. **Primary source tail dichotomy/exhaustion.** The fixed-dilation and bounded macroscopic-parent tail cases are controlled; genuinely nonlocal parents must be source-accounted or routed to charge/leakage.
3. **K3 quotient-or-charge graph theorem.** Prove the finite weighted graph statement (7), including the oscillation coordinate Φ_{osc} .
4. **K4 forward absorption.** Prove

$$P\mathcal{F}_{\text{phys}}^{\text{final}} \leq C_{\text{forw}}^{\text{final}}D + o.$$

5. **K5B high-strain matrix rigidity.** Prove that source-weighted high strain on a nonrenewing, nonseparating branch core is paid by shape, oscillation/twist, renewal, or separation, with the source-weighted path normalization and the constant +1 routed to P .
6. **Packed boundary reverse constants.** The packed boundary row closes good-scale null-source regularity. K5 still needs reverse-row constants for the same packed row.
7. **K6 spectral and loop-gap certification.** Supply actual weights w and constants with $q_A(w) < 1$ and

$$C_{\text{forw}}^{\text{final}}C_0 < c_{\text{final}}(\eta).$$

8. **K7 compatibility under ledger edits.** Any change to $D(r)$ must preserve Proposition 1.

10 Conclusion

The proposed law is deliberately simple:

$$\boxed{\text{Amplification} \leq \text{Payment}.}$$

This manuscript defines the payment ledger, states the conditional theorem, records the partial closures, reports adversarial mechanism tests, and names the gates that remain. It is a research program paper, not a completed Clay proof. Its usefulness is that it turns a vague blowup question into a finite set of attackable accounting failures. The next stage is therefore not to reinterpret the closure theorem, but to prove or falsify the named analytic gates K3–K7 with explicit constants.

References

- [1] J. Leray, Sur le mouvement d'un liquide visqueux emplissant l'espace, *Acta Math.* **63** (1934), 193–248. doi:10.1007/BF02547354.
- [2] E. Hopf, Über die Anfangswertaufgabe für die hydrodynamischen Grundgleichungen, *Math. Nachr.* **4** (1951), 213–231.
- [3] G. Prodi, Un teorema di unicità per le equazioni di Navier–Stokes, *Ann. Mat. Pura Appl.* **48** (1959), 173–182.
- [4] J. Serrin, On the interior regularity of weak solutions of the Navier–Stokes equations, *Arch. Rational Mech. Anal.* **9** (1962), 187–195. doi:10.1007/BF00253344.
- [5] L. Caffarelli, R. Kohn, and L. Nirenberg, Partial regularity of suitable weak solutions of the Navier–Stokes equations, *Comm. Pure Appl. Math.* **35** (1982), 771–831. doi:10.1002/cpa.3160350604.
- [6] J. T. Beale, T. Kato, and A. Majda, Remarks on the breakdown of smooth solutions for the 3-D Euler equations, *Comm. Math. Phys.* **94** (1984), 61–66. doi:10.1007/BF01212349.
- [7] P. Constantin and C. Fefferman, Direction of vorticity and the problem of global regularity for the Navier–Stokes equations, *Indiana Univ. Math. J.* **42** (1993), 775–789. doi:10.1512/iumj.1993.42.42034.
- [8] F.-H. Lin, A new proof of the Caffarelli–Kohn–Nirenberg theorem, *Comm. Pure Appl. Math.* **51** (1998), 241–257. doi:10.1002/(SICI)1097-0312(199803)51:3%3C241::AID-CPA2%3E3.0.CO;2-A.
- [9] L. Escauriaza, G. A. Seregin, and V. Šverák, $L^{3,\infty}$ -solutions of Navier–Stokes equations and backward uniqueness, *Russian Math. Surveys* **58** (2003), 211–250. doi:10.1070/RM2003v058n02ABEH000609.
- [10] Y. Li et al., A public turbulence database cluster and applications to study Lagrangian evolution of velocity increments in turbulence, *Journal of Turbulence* **9** (2008), N31. doi:10.1080/14685240802376389.
- [11] E. Perlman, R. Burns, Y. Li, and C. Meneveau, Data exploration of turbulence simulations using a database cluster, in *SC07: Proceedings of the 2007 ACM/IEEE Conference on Supercomputing*, 2007. doi:10.1145/1362622.1362654.

- [12] T. Y. Hou, Blow-up or no blow-up? A unified computational and analytic approach to 3D incompressible Euler and Navier–Stokes equations, *Acta Numerica* **18** (2009), 277–346. doi:10.1017/S0962492906420018.